

Data and descriptive Statistics

“Errors using inadequate data are much less than those using no data at all.”

Charles Babbage

Working memory

We can remember about four things only

Usual to use **repetition, nesting/pattern generation** in order to boost this

Actually better than you might expect – consider how many physical objects we can handle at once...

Solution

We need to characterise the data with fewer parameters – ideally the 3 or 4 we can actually work with at one time

The average:

What number is typical of our data?

The range:

What are the outer limits of our data?

The variance:

how evenly spread out is our data?

Averages

Three main terms to use as average:

Mean - **Arithmetic average**

Median - **Middle value**

Mode - **Most common value**

Mean

Arithmetic average

Sum all data values and divide by number of data points

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

What is the average wage in the U.K. ?

Find n and find $\sum x$

Pop. is approximately* **65,000,000**

*Effectively unknowable

GDP

Gross Domestic Product is about:

\$2500,000,000,000

Finding the mean we take 2.5 trillion, divide by 65 million and get:

\$38500

Accounting for the exchange rate we get:

£24,000

Median

Line all your data point up in order from smallest to largest – the one in the middle is the median

$$x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ \dots \ x_n$$

$$\mathbf{M} = x_{(n+1)/2}$$

Can give a fractional position!

Median

[1, 2, 3, 4, 5]

Median = 3

[1, 2, * 3, 4]

*Median = $\frac{2+3}{2} = 2.5$

Quartiles

Divide our data into quarters, lower quartile calculated from the lowest point, upper quartile from the highest

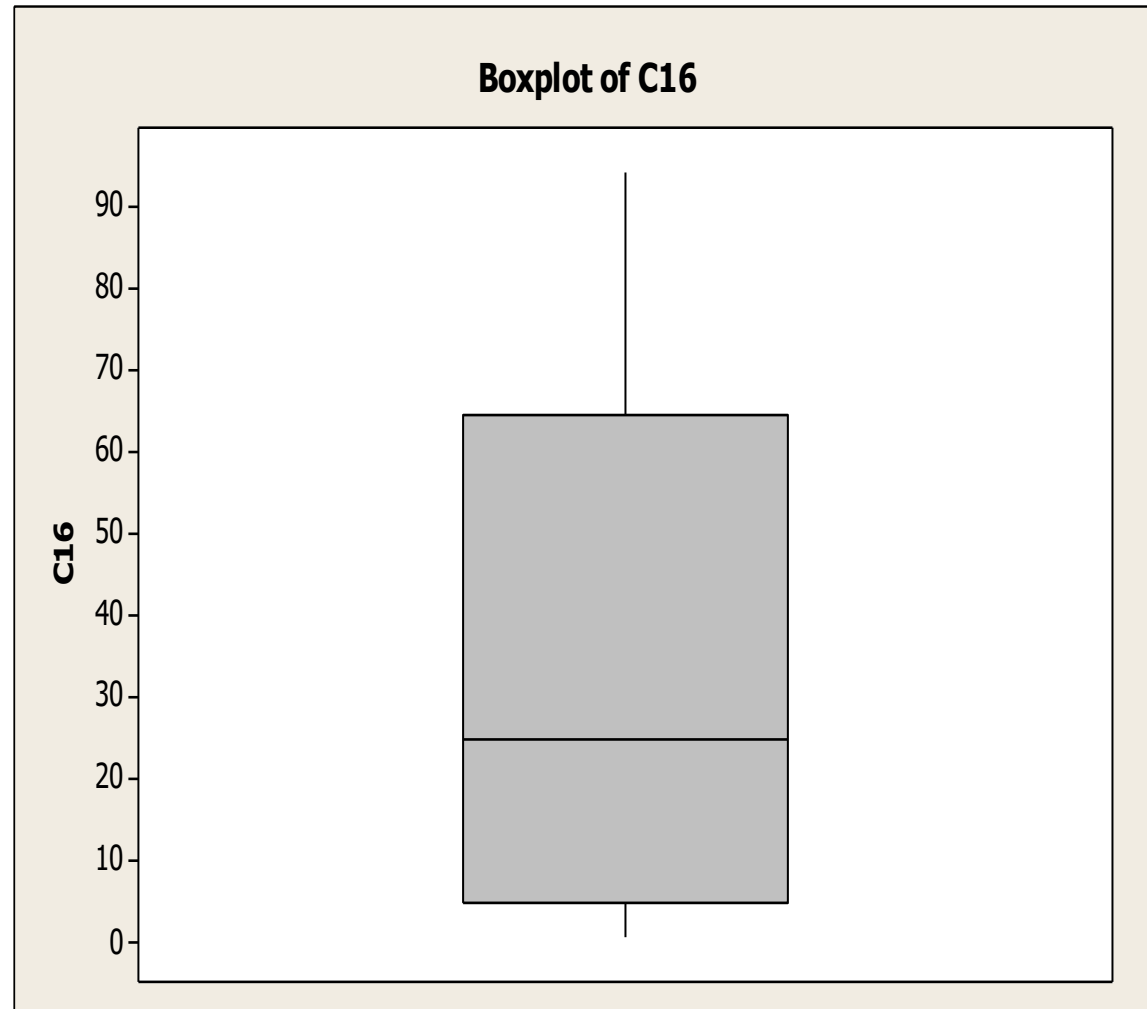
Sensible to use for large datasets

Quartile position is: $\frac{\text{median} + 1}{2}$

Something is wrong with my title and axis label here – what is it???

Box and Whisker plot

Relies on a combination of the range, median and quartiles to visually characterise a dataset



1
2
3
5
5
6
8
18
25
34
47
56
60
69
71
88
94

Ranges

1

2

3

5

5

6

8

18

25

34

47

56

60

69

71

88

94

Range = Largest minus smallest: $94 - 1 = 93$

Interquartile range = Range of the two central quartiles: $60 - 5 = 55$

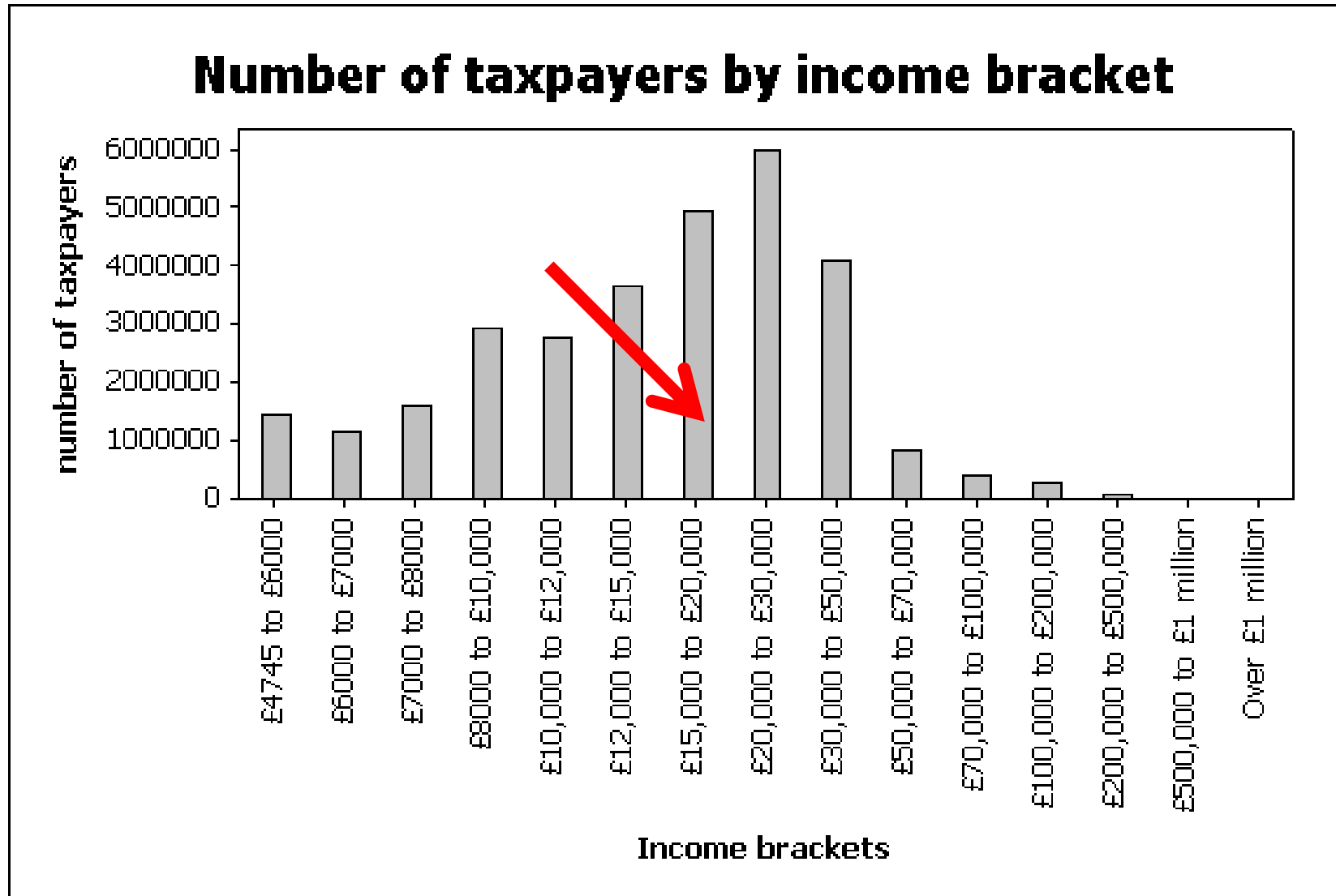
Semi interquartile range = Half the interquartile range: $55 / 2 = 27.5$

From HMRC

range	number of taxpayers
£4745 to £6000	1,440,000
£6000 to £7000	1,160,000
£7000 to £8000	1,590,000
£8000 to £10,000	2,950,000
£10,000 to £12,000	2,760,000
£12,000 to £15,000	3,650,000
£15,000 to £20,000	4,950,000
£20,000 to £30,000	6,000,000
£30,000 to £50,000	4,090,000
£50,000 to £70,000	859,000
£70,000 to £100,000	410,000
£100,000 to £200,000	300,000
£200,000 to £500,000	89,000
£500,000 to £1 million	16,000
Over £1 million	6,000

Something is wrong with my choice of graph here – what is it???

Median = £16,400



Mode

The most commonly occurring value

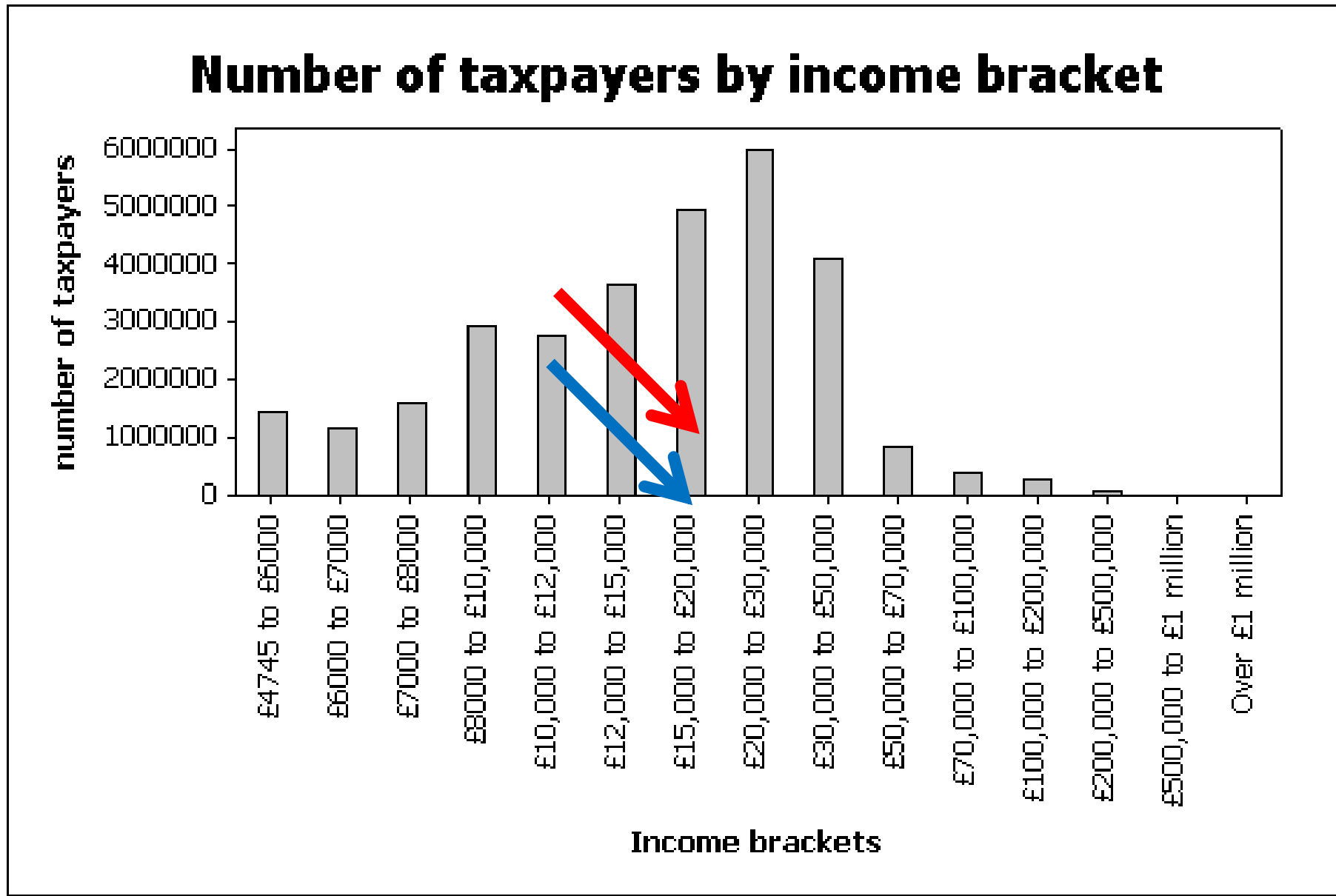
No convenient mathematical formula

We have to bracket the income for it to be meaningful

For large data sets usually found through graphical display (dotplot or similar)

Something is wrong with my use of brackets here – what is it???

Mode = £15,000



Average wage in the U.K. ?

Mean = 24, 000

Median= 16,400

Mode = £15,000

Where is our mistake?

Disparity in measures due to:

Non-normal population

Variations in geographical income

No account taken of part time work (pro-rata)

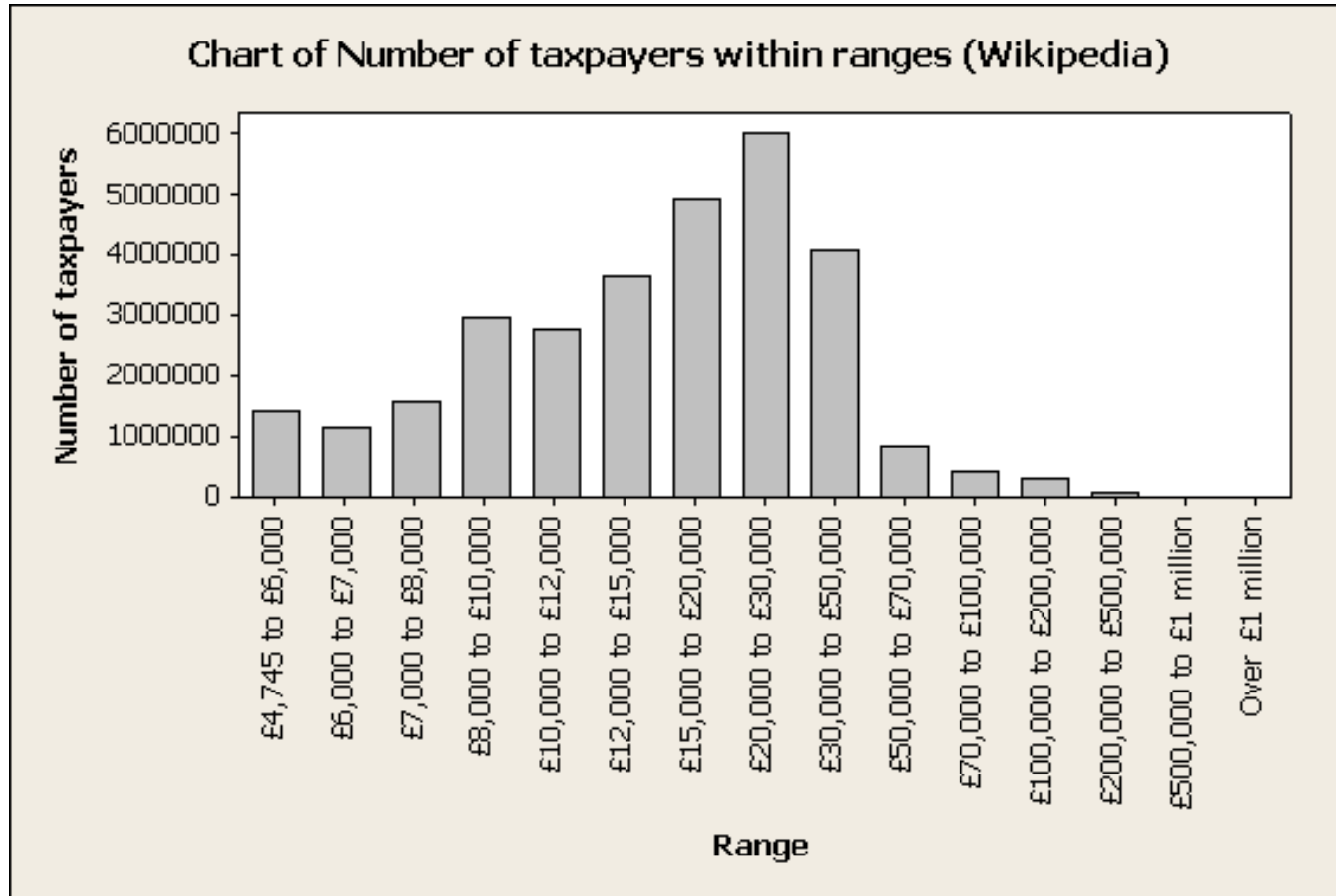
Sick/maternity pay less than normal

Self employed people self report!

Skewed by a few mega-rich individuals

Something is wrong with my source for this information – what is it???

Taxpayer issues



Tax evasion, avoidance

Off shore earnings

Income tax or capital gains tax or,
corporation tax or dividends tax?

Which is best?

Median usually taken as most accurate (less subject to outliers) when looking at non-normal data

Mode usually ignored (tends to be close to median)

Mean often used to prove a point in financial stats – this is usually best with Normal data

Samples and variance

In the last example we had some estimations (population) but we also had a complete count of some data (HMRC)

We knew the information for the whole population

Usually this isn't the case, usually we only have a sample

This class can be treated as a sample of the NTU student population

In some aspects there will be very little variation (number of **noses**, **Height in m**, etc.)

In some aspects there will be much more variation (number of **freckles**, number of **tattoos**, length of **hair in cm** etc.)

Deviation and the Mean

Take the simple data set: [1,2,3,4]

The mean is found by taking the sum of the values and dividing by the amount of data points we have:

$$M=(1+2+3+4)/4$$

$$M=10/4$$

$$M=2.5$$

Even though the 'average' value for this data set is **2.5**, none of the data points takes that value exactly.

The data set [-1000, 608, 400, 2] also has a mean of **2.5**

We know how accurate our sample is likely to be by looking at how far from the mean each of the values is. This distance from the mean is called the ***deviation***

Finding deviation

$$1 - 2.5 = -1.5$$

$$2 - 2.5 = -0.5$$

$$3 - 2.5 = 0.5$$

$$4 - 2.5 = 1.5$$

We can find the average distance of the data points from the mean the same way we find any average – add them up and divide by however many there are.

The total **distance** for the first two points is 2, the total for the second two points is also 2. Only the **sign** varies

In this (& every other) case they sum to zero.

To avoid this we **square** the values

Doing so make them all positive so no zero-sums

We add up all the squared values (giving the '**sum of squares**')
sum of squares

The bigger the 'sum of squares' the more varied the data.

Brilliant, problem solved! 😊

$$1 - 2.5 = -1.5 \quad (-1.5)^2 = 2.25$$

$$2 - 2.5 = -0.5 \quad (-0.5)^2 = 0.25$$

$$3 - 2.5 = 0.5 \quad (0.5)^2 = 0.25$$

$$4 - 2.5 = 1.5 \quad (1.5)^2 = 2.25$$

$$\text{sum of squares} = 5$$

Problem not solved 😞

Consider this data set: **[1, 1, 2, 2, 3, 3, 4, 4]**

It is twice as big as the last one and the end sum is twice as big also. Can this be true?

Clearly this set does not vary more than the first - so its deviation should be the same

$$1 - 2.5 = -1.5 \quad (-1.5)^2 = 2.25$$

$$1 - 2.5 = -1.5 \quad (-1.5)^2 = 2.25$$

$$2 - 2.5 = -0.5 \quad (-0.5)^2 = 0.25$$

$$2 - 2.5 = -0.5 \quad (-0.5)^2 = 0.25$$

$$3 - 2.5 = 0.5 \quad (0.5)^2 = 0.25$$

$$3 - 2.5 = 0.5 \quad (0.5)^2 = 0.25$$

$$4 - 2.5 = 1.5 \quad (1.5)^2 = 2.25$$

$$4 - 2.5 = 1.5 \quad (1.5)^2 = 2.25$$

sum=10

We need to account for the fact that bigger groups will have a higher sum

To do so we divide by **n**

This gives us:

$(\text{sum of } 5) / (4 \text{ values}) = 1.25$ for the first dataset

$10/8 = 1.25$ for the second, much better!

Variance

The quantity we have calculated is called the Variance of the data

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

We write the variance as σ^2 because we often find σ more useful.

What about the units though – does it matter if we measure height in m or cm?

[10, 20, 30, 40]

$$10-25=-15 \quad (-15)^2 = 225$$

$$20-25=-5 \quad (-5)^2 = 25$$

$$30-25=5 \quad (5)^2 = 25$$

$$40-25=15 \quad (15)^2 = 225$$

Sum of squares = 500 sum/n=125

Does this data set really vary **100 times** more than [1, 2, 3, 4]?

The range fallacy

Consider this data set: **[1.5, 1.5, 3.5, 3.5]**

It also has **n=4**

It also has **mean=2.5**

$$1.5 - 2.5 = -1 \quad (-1)^2 = 1$$

$$1.5 - 2.5 = -1 \quad (-1)^2 = 1$$

$$3.5 - 2.5 = 1 \quad (1)^2 = 1$$

$$3.5 - 2.5 = 1 \quad (1)^2 = 1$$

$$\text{sum}=4, \text{sum}/n = 1$$

$$1 - 2.5 = -1.5 \quad (-1.5)^2 = 2.25$$

$$2 - 2.5 = -0.5 \quad (-0.5)^2 = 0.25$$

$$3 - 2.5 = 0.5 \quad (0.5)^2 = 0.25$$

$$4 - 2.5 = 1.5 \quad (1.5)^2 = 2.25$$

$$\text{sum}=5, \text{sum}/n = 1.25$$

Is this always true?

We have the same measure of variation but we have different datasets. One varies from 1.5-3.5, the other from 1-4.

Isn't it only natural and correct that the group with the wider **range** is considered to vary more?

In the absence of other info, it is fine to expect a larger range to mean a larger variance but it isn't guaranteed!

[1,1,1,1,1,1,1,1,1,1,1,1,1,5]

This group has an even wider range so should have a greater deviation value – in fact it has a variance slightly **less** than the group [1,2,3,4]

The greater n compensates for the greater range in this case.

Variance or deviation?

The **variance** of a population is:

$$\sigma^2 = \frac{\sum_{x_1}^{x_N} (x_i - \mu)^2}{N}$$

μ is the population mean

The **standard deviation** of a population is:

$$\sigma = \sqrt{\frac{\sum_{x_1}^{x_N} (x_i - \mu)^2}{N}}$$

Sample or population

Sample uses English lettering

s = standard deviation,

$\bar{x}$ = mean of x-values

n = size of sample (number of x values)

Population uses *mostly* Greek lettering

σ = sigma = standard deviation

μ = mu = mean

N = size of population

Pop Vs. Sample standard deviation

$$\sigma = \sqrt{\frac{\sum_{x_1}^x \textcolor{teal}{N} (x_i - \textcolor{red}{\mu})^2}{N}}$$

$$s = \sqrt{\frac{\sum_{x_1}^x \textcolor{teal}{n} (x_i - \textcolor{red}{\bar{x}})^2}{n-1}}$$

n-1?

$$s = \sqrt{\frac{\sum_{x_1}^{x_n} (x_i - \bar{x})^2}{n - 1}}$$

Called **Bessel's correction**

The values in a sample are likely to be closer to the mean of the **sample** than the mean of the **population**

We usually want to use the sample as an approximation for the population
Reducing n makes the variance value bigger

It has a greater effect on smaller samples than larger samples

Relates to the **degrees of freedom**

Finally, break it down!

First find the **mean** of all the x values

Find $(x - \text{mean})$ for each x value

Square these: $(x - \text{mean})(x - \text{mean})$

Add all the squared values together

Divide by (the number of values $- 1$)

Take the square root

$$s = \sqrt{\frac{\sum_{x_1}^{x_n} (x_i - \bar{x})^2}{n - 1}}$$

Summary

Mean, median and **mode** are types of average – they can be different for a single data set and they are best used for different things

Range, interquartile range and semi-interquartile range

Standard deviation and **Variance** are measures of how closely together your data is clustered – they are calculated differently for **samples** and **populations**

Terminology & sources

Mean

Median

Mode

Quartiles

Range

Interquartile range

Semi-interquartile range

Variance

Standard deviation

Bessel's correction

Degrees of freedom

Population

Sample

[More info on memory here](#)

[Income data here](#)

p388-402 in Foundation Maths by
Tony Croft